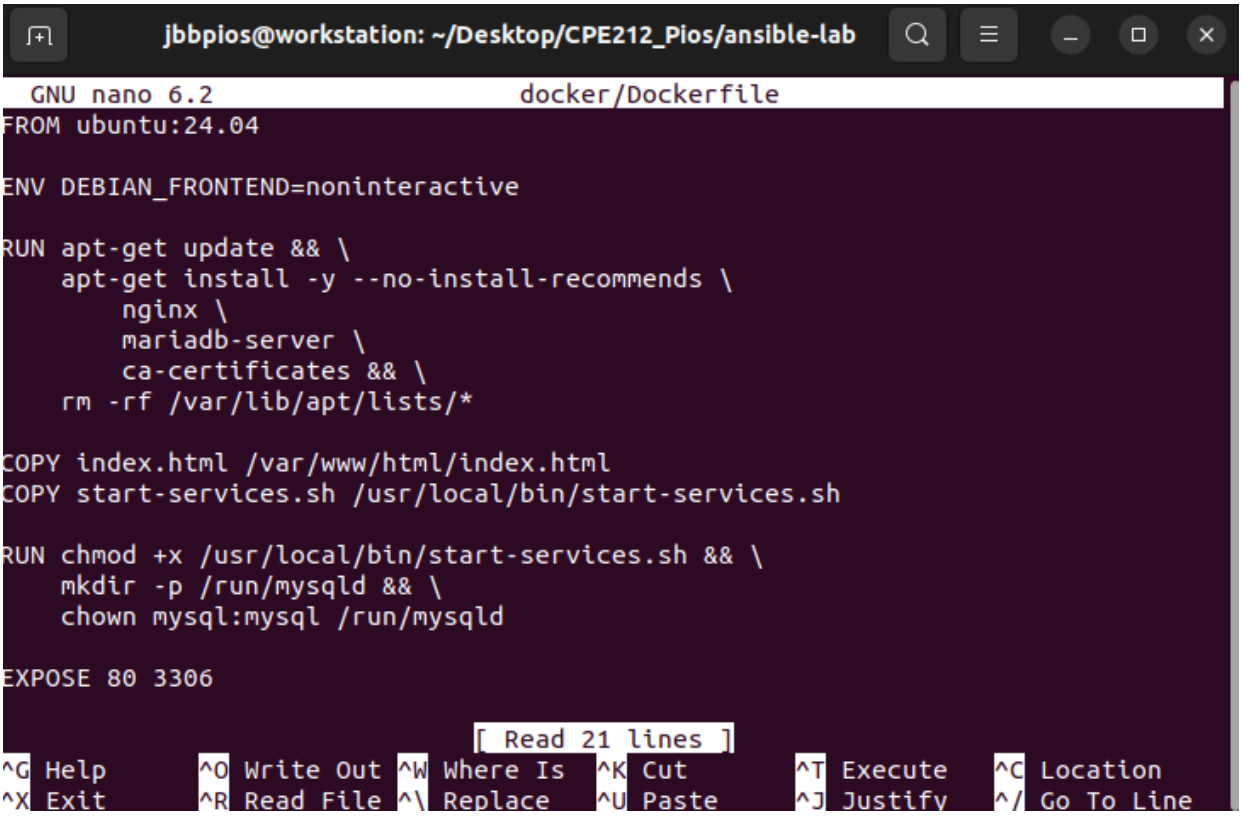
COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	October 13, 2026	**DATE SUBMITTED:**	October 13, 2026
NAME:	Pios, Joshua Paul B.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. 11.1 **Containerization**

1. PROCEDURE OUTPUT



jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab

```
 GNU nano 6.2                        docker/index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>CPE212 Activity 11</title>
</head>
<body>
    <h1>Activity 11: Containerization</h1>
    <p>Nginx is running inside a Docker container.</p>
    <p>This image was built using Ansible automation.</p>
</body>
</html>

                          [ Read 13 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab

```
 GNU nano 6.2                   docker/start-services.sh
#!/bin/bash

set -e

mkdir -p /run/mysqld
chown mysql:mysql /run/mysqld

if [ ! -d /var/lib/mysql/mysql ]; then
    mariadb-install-db --user=mysql --datadir=/var/lib/mysql
fi

mariadbd --user=mysql --console &
DB_PID=$!

for i in $(seq 1 30); do
    if mariadb-admin ping --silent; then
        break
    fi
    sleep 1
done

                          [ Read 37 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ cat ansible.cfg
[defaults]
inventory = inventory/hosts
host_key_checking = False
interpreter_python = auto_silent
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$
```

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ cat inventory/host
s
all:
  children:

    elasticsearch:
      hosts:
        server1:
          ansible_host: 192.168.56.101
          ansible_user: jppios

    kibana:
      hosts:
        server2:
          ansible_host: 192.168.56.102
          ansible_user: jpbpios

    logstash:
      hosts:
        server3:
          ansible_host: 192.168.56.104
          ansible_user: jpios
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$
```

jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab

```
 GNU nano 6.2                          activity11.yml
---
- name: Install and configure Docker on all VMs
  hosts: all
  become: true
  gather_facts: true

  tasks:
    - name: Install prerequisites on Ubuntu
      ansible.builtin.apt:
        name:
          - ca-certificates
          - curl
        state: present
        update_cache: true
      when: ansible_os_family == "Debian"

    - name: Create Docker repository key directory on Ubuntu
      ansible.builtin.file:
        path: /etc/apt/keyrings
        state: directory
                              [ Read 105 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-playbook a
ctivity11.yml --syntax-check

playbook: activity11.yml
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-playbook a
ctivity11.yml -K
BECOME password:

PLAY [Install and configure Docker on all VMs] *********************************

TASK [Gathering Facts] *********************************************************
ok: [server2]
ok: [server1]
ok: [server3]

TASK [Install prerequisites on Ubuntu] *****************************************
skipping: [server3]
ok: [server2]
ok: [server1]

TASK [Create Docker repository key directory on Ubuntu] ************************
skipping: [server3]
```

```
ok: [server1]
ok: [server2]
changed: [server3]

TASK [Add the Ansible SSH user to the Docker group] ****************************
changed: [server1]
changed: [server2]
changed: [server3]

TASK [Verify Docker installation] **********************************************
ok: [server1]
ok: [server2]
ok: [server3]

PLAY RECAP *********************************************************************
server1                    : ok=10   changed=4    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
server2                    : ok=10   changed=4    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
server3                    : ok=8    changed=5    unreachable=0    failed=0    s
kipped=5    rescued=0    ignored=0
```

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible all -m com
mand -a "docker --version"
server2 | CHANGED | rc=0 >>
Docker version 29.9.0, build f415da8
server1 | CHANGED | rc=0 >>
Docker version 29.9.0, build f415da8
server3 | CHANGED | rc=0 >>
Docker version 29.9.0, build f415da8
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible all -b -K
-m command -a "systemctl is-active docker"
BECOME password:
server1 | CHANGED | rc=0 >>
active
server2 | CHANGED | rc=0 >>
active
server3 | CHANGED | rc=0 >>
active
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

What are the benefits of implementing containerizations?

Containerization is beneficial because it makes applications easier to deploy, manage, and run on different systems. It also helps avoid compatibility issues because the application and its dependencies are packaged together. With Docker, applications can run in isolated environments while using fewer resources than traditional virtual machines.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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